

Sysmon + Wazuh Integration Lab Guide

This lab helps students monitor Windows security events using **Sysmon** and **Wazuh**, correlating them with threat intel via **MISP**. It's designed for practical cybersecurity learning with alert generation and visualization.

What is Sysmon?

Sysmon (System Monitor) is a Windows system service and device driver from Sysinternals that logs system activity to the Windows Event Log. It captures events like:

- Process creation
- Network connections
- File modifications
- Registry changes
- DNS queries

 Unlike **Event Viewer**, which logs all Windows logs (noisy and broad), Sysmon:

- Focuses on security-relevant **behavioral logs**
 - Enables **fine-tuned detection** using custom rules
 - Is essential for **SOC** operations & **threat hunting**
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STEP 1 – Install Sysmon on Windows Agent

1. Download Sysmon:
 - <https://docs.microsoft.com/en-us/sysinternals/downloads/sysmon>
2. Extract sysmon.zip and download the config from:
 - <https://github.com/SwiftOnSecurity/sysmon-config>
3. Save the config as: sysmonconfig-export.xml
4. Launch PowerShell as **Administrator** and run:

To install:

```
./sysmon64.exe -accepteula -i sysmonconfig-export.xml
```

If already installed:

```
./sysmon64.exe -u # uninstall
```

```
./sysmon64.exe -i sysmonconfig-export.xml # reinstall
```

✅ Result: Logs will start appearing under: Microsoft-Windows-Sysmon/Operational

⚙️ STEP 2 – Configure Wazuh Agent to Monitor Sysmon

On the **Windows Agent** system:

1. Edit Wazuh agent config file:

```
<localfile>
```

```
<location>Microsoft-Windows-Sysmon/Operational</location>
```

```
<log_format>eventchannel</log_format>
```

```
</localfile>
```

2. Restart Wazuh Agent:

```
Restart-Service wazuh
```

🔧 STEP 3 – Change Rule Level for DNS Event (ID 22)

On **Wazuh Manager**:

1. Edit the existing rule file (do **not** use local_rules.xml):

```
sudo nano /var/ossec/ruleset/rules/0950-sysmon_id_20.xml
```

2. Modify or add the rule (if not already there):

```
<rule id="61650" level="8" overwrite="yes">
```

```
<if_sid>61600</if_sid>
```

```
<field name="win.system.eventID">^22$</field>
```

```
<description>Sysmon - Event ID 22: DNSEvent (DNS query)</description>
```

```
<options>no_full_log</options>
```

```
<group>sysmon_event_22,</group>

</rule>
```

3. Restart Wazuh Manager:

```
sudo systemctl restart wazuh-manager
```

STEP 4 – Test Sysmon Logging

On the **Windows agent**, try:

```
nslookup facebook.com
```

Then, on **Wazuh Manager**, check logs:

```
tail -f /var/ossec/logs/ossec.log
```

You should see DNS queries appear.

STEP 5 – View in Wazuh Dashboard

1. Go to Wazuh Dashboard (default: <https://:5601>)

2. Navigate to:

- **Security Events → Sysmon Event 22**
- Use search:

```
rule.description:"Sysmon - Event ID 22: DNSEvent"
```

Other Use Cases to Monitor

Besides DNS, you can monitor:

- **Process Creation (ID 1)** → Detect suspicious programs
- **Network Connect (ID 3)** → See external connections
- **FileCreateTime (ID 2)** → Timestamping
- **Registry Edits (ID 13)** → Persistence techniques